

SOLIDWORKS API (.NET VBA C#) — Reference

(261/337): ISimulationForceFeatureData- ISimulationLinearSpringFeatureData

SOLIDWORKS API Documentation (converted from CHM)

ISimulationForceFeatureData (Type)

ISimulationForceFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationForceFeatureData Interface

Allows access to a force or torque feature in SOLIDWORKS Motion studies.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISimulationForceFeatureData
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
```

C#

```
public interface ISimulationForceFeatureData
```

C++/CLI

```
public interface class ISimulationForceFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData](#).

Example

[Create Force Feature \(VBA\)](#)
[Fire Events for External Output Time Step Changes \(C#\)](#)
[Fire Events for External Output Time Step Changes \(VB.NET\)](#)
[Fire Events for External Output Time Step Changes \(VBA\)](#)

Accessors

[IFeature::GetDefinition](#) and [IFeature::!GetDefintion](#)
[IMotionStudy::CreateFeature](#)

Access Diagram

[SimulationForceFeatureData](#)

See Also

[ISimulationForceFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISimulationGravityFeatureData Interface](#)
[ISimulationMotorFeatureData Interface](#)
[Simulation3DContactFeatureData Interface](#)
[SimulationDamperFeatureData Interface](#)
[ISimulationSpringFeatureData Interface](#)

ISimulationForceFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationForceFeatureData Interface

The following tables list the members exposed by [ISimulationForceFeatureData](#).

Public Properties

Name	Description
 ActionDirection	Gets or sets the direction of the force.
 ActionLocation	Gets or sets the location at which to apply the force for an action-only force.
 ActionType	Gets or sets the type of action for this Force feature.
 ExternalState	Gets or sets whether your application can listen to force-related motion study events.
 ForceFunctionType	Gets or sets the type of function for this Force feature.
 ForceType	Gets the type of force. NOTE: This property is a get-only property. Set is not implemented.
 FunctionConstantValue	Gets or sets the function constant value for this Force feature.
 FunctionExpression	Gets or sets the expression function for this Force feature.
 FunctionInterpolatedValues	Gets or sets the interpolated values for this Force feature.
 InterpolationScheme	Gets the interpolation scheme for this Force feature.
 LoadReferences	Gets or sets the load references for this Force feature.
 ReactionLocation	Gets or sets the location at which to apply the force for an action/reaction force.
 ReferenceComponent	Gets or sets the component to serve as a reference frame for the force.
 ReverseDirection	Gets or sets whether to reverse the direction of the force.

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Public Methods

Name	Description
 GetEndPoints	Gets the end points of this Force feature.
 GetFunctionHarmonicValues	Gets the harmonic function values for this Force feature.
 GetFunctionStepValues	Gets the step function, whose magnitude transitions smoothly from one value to another value, for this Force feature.
 GetInterpolatedValue	Gets the interpolated value at the specified time for this Force feature.
 LoadSplineData	Loads the spline data from the specified file for this Force feature.
 SetEndPoints	Sets the end points for this Force feature.
 SetFunctionHarmonicValues	Sets the harmonic function values for this Force feature.
 SetFunctionStepValues	Sets the step function, whose magnitude transitions smoothly from one value to another value, for this Force feature.

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See Also

[ISimulationForceFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISimulationGravityFeatureData Interface](#)
[ISimulationMotorFeatureData Interface](#)
[Simulation3DContactFeatureData Interface](#)
[SimulationDamperFeatureData Interface](#)
[ISimulationSpringFeatureData Interface](#)

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ISimulationForceFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationForceFeatureData Interface

For a list of all members of this type, see [ISimulationForceFeatureData members](#).

Public Methods

Name	Description
 GetEndPoints	Gets the end points of this Force feature.

	GetFunctionHarmonicValues	Gets the harmonic function values for this Force feature.
	GetFunctionStepValues	Gets the step function, whose magnitude transitions smoothly from one value to another value, for this Force feature.
	GetInterpolatedValue	Gets the interpolated value at the specified time for this Force feature.
	LoadSplineData	Loads the spline data from the specified file for this Force feature.
	SetEndPoint	Sets the end points for this Force feature.
	SetFunctionHarmonicValues	Sets the harmonic function values for this Force feature.
	SetFunctionStepValues	Sets the step function, whose magnitude transitions smoothly from one value to another value, for this Force feature.

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See Also

[ISimulationForceFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISimulationGravityFeatureData Interface](#)
[ISimulationMotorFeatureData Interface](#)
[Simulation3DContactFeatureData Interface](#)
[SimulationDamperFeatureData Interface](#)
[ISimulationSpringFeatureData Interface](#)

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ISimulationForceFeatureData Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : [ISimulationForceFeatureData Interface](#)

For a list of all members of this type, see [ISimulationForceFeatureData members](#).

Public Properties

Name	Description
 ActionDirection	Gets or sets the direction of the force.
 ActionLocation	Gets or sets the location at which to apply the force for an action-only force.
 ActionType	Gets or sets the type of action for this Force feature.
 ExternalState	Gets or sets whether your application can listen to force-related motion study events.
 ForceFunctionType	Gets or sets the type of function for this Force feature.
 ForceType	Gets the type of force.
NOTE: This property is a get-only property. Set is not implemented.	
 FunctionConstantValue	Gets or sets the function constant value for this Force feature.
 FunctionExpression	Gets or sets the expression function for this Force feature.
 FunctionInterpolatedValues	Gets or sets the interpolated values for this Force feature.
 InterpolationScheme	Gets the interpolation scheme for this Force feature.
 LoadReferences	Gets or sets the load references for this Force feature.
 ReactionLocation	Gets or sets the location at which to apply the force for an action/reaction force.
 ReferenceComponent	Gets or sets the component to serve as a reference frame for the force.
 ReverseDirection	Gets or sets whether to reverse the direction of the force.

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See Also

[ISimulationForceFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISimulationGravityFeatureData Interface](#)
[ISimulationMotorFeatureData Interface](#)
[Simulation3DContactFeatureData Interface](#)
[SimulationDamperFeatureData Interface](#)
[ISimulationSpringFeatureData Interface](#)

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ActionDirection Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : ActionDirection Property (ISimulationForceFeatureData)

Gets or sets the direction of the force.

.NET Syntax

Visual Basic (Declaration)

```
Property ActionDirection As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim value As System.Object
```

```
instance.ActionDirection = value
```

```
value = instance.ActionDirection
```

C#

```
System.object ActionDirection {get; set;}
```

C++/CLI

```
property System.Object^ ActionDirection {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

- Along an axis defined by an edge or a plane
- Along the line-of-sight between two points

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ActionDirection](#).

Remarks

The way you define the directions of your forces depends on which forces you are creating. If you are creating an action-only force, the force direction remains fixed with respect to some part in your model, either a moving part or the ground part. In this case, you can define the force direction using one vector defined by an edge or a plane.

If you are creating an action/reaction force, then the direction along which you want the force applied is defined by the line between two points in your model and is constantly changing throughout the simulation. In this case, you can define the force direction as the line between the two points.

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::ActionLocation Property](#)
[ISimulationForceFeatureData::ActionType Property](#)
[ISimulationForceFeatureData::ReverseDirection Property](#)
[ISimulationForceFeatureData::ReactionLocation Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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ActionLocation Property (ISimulationForceFeatureData)

[See Also](#) [Example](#)



Gets or sets the location at which to apply the force for an action-only force.

.NET Syntax

Visual Basic (Declaration)

Property ActionLocation As System.Object

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim value As System.Object
```

```
instance.ActionLocation = value
```

```
value = instance.ActionLocation
```

C#

```
System.Object ActionLocation {get; set;}
```

C++/CLI

```
property System.Object^ ActionLocation {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Vertex, circular edge, or linear edge to define the points at which the moment is applied

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ActionLocation](#).

Example

[Create Force Feature \(VBA\)](#)

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::ActionDirection Property](#)
[ISimulationForceFeatureData::ActionType Property](#)
[ISimulationForceFeatureData::ReactionLocation Property](#)

Availability

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ActionType Property (ISimulationForceFeatureData)

[See Also](#) [Example](#)



Gets or sets the type of action for this Force feature.

.NET Syntax

Visual Basic (Declaration)

Property ActionType As System.Integer

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim value As System.Integer

instance.ActionType = value
value = instance.ActionType
```

C#

```
System.int ActionType {get; set;}
```

C++/CLI

```
property System.int ActionType {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of action as defined in [swSimulationForceActionType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ActionType](#).

Example

[Create Force Feature \(VBA\)](#)

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::ActionDirection Property](#)
[ISimulationForceFeatureData::ActionLocation Property](#)
[ISimulationForceFeatureData::ReactionLocation Property](#)

Availability

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ExternalState Property (ISimulationForceFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : ExternalState Property (ISimulationForceFeatureData)

Gets or sets whether your application can listen to force-related motion study events.

.NET Syntax

Visual Basic (Declaration)

```
Property ExternalState As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim value As System.Boolean

instance.ExternalState = value
value = instance.ExternalState
```

C#

```
System.bool ExternalState {get; set;}
```

C++/CLI

```
property System.bool ExternalState {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to listen to [IMotionStudy.ForceTimeStampChangeNotify](#) and [ForceOutputTimeStampChangeNotify](#) events, false to not

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ExternalState](#).

Example

[Fire Events for External Output Time Step Changes \(C#\)](#)
[Fire Events for External Output Time Step Changes \(VB.NET\)](#)
[Fire Events for External Output Time Step Changes \(VBA\)](#)

Remarks

When a force is set to external, IMotionStudy.ForceTimeStampChangeNotify and ForceOutputTimeStampChangeNotify events are called with the motor name, time, and a pointer for the return value. These events are called at every major time step from ADAMS. You can implement any force law after catching these events. The return value for each event is a double that specifies the new force or torque magnitude.

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)

Availability

SOLIDWORKS 2008 SP1, Revision Number 16.1

SOLIDWORKS API Help

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ForceFunctionType Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : ForceFunctionType Property (ISimulationForceFeatureData)

Gets or sets the type of function for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Property ForceFunctionType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim value As System.Integer
```

```
instance.ForceFunctionType = value
```

```
value = instance.ForceFunctionType
```

C#

```
System.int ForceFunctionType {get; set;}
```

C++/CLI

```
property System.int ForceFunctionType {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of function as defined in [swSimulationForceFunctionType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ForceFunctionType](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::GetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::GetFunctionStepValues Method](#)
[ISimulationForceFeatureData::SetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::SetFunctionStepValues Method](#)
[ISimulationForceFeatureData::FunctionConstantValue Property](#)
[ISimulationForceFeatureData::FunctionExpression Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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ForceType Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : ForceType Property (ISimulationForceFeatureData)

Gets the type of force.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ForceType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim value As System.Integer

value = instance.ForceType
```

C#

```
System.int ForceType {get;}
```

C++/CLI

```
property System.int ForceType {
    System.int get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of force as defined in [swSimulationForceType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ForceType](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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FunctionConstantValue Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : FunctionConstantValue Property (ISimulationForceFeatureData)

Gets or sets the function constant value for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Property FunctionConstantValue As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim value As System.Double
```

```
instance.FunctionConstantValue = value
```

```
value = instance.FunctionConstantValue
```

C#

```
System.double FunctionConstantValue {get; set;}
```

C++/CLI

```
property System.double FunctionConstantValue {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Function constant value

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::FunctionConstantValue](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::GetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::GetFunctionStepValues Method](#)
[ISimulationForceFeatureData::SetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::SetFunctionStepValues Method](#)
[ISimulationForceFeatureData::FunctionExpression Property](#)
[ISimulationForceFeatureData::ForceFunctionType Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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FunctionExpression Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : FunctionExpression Property (ISimulationForceFeatureData)

Gets or sets the expression function for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Property FunctionExpression As System.String
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim value As System.String
```

```
instance.FunctionExpression = value
```

```
value = instance.FunctionExpression
```

C#

```
System.string FunctionExpression {get; set;}
```

C++/CLI

```
property System.String^ FunctionExpression {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Expression function in document units

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::FunctionExpression](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::GetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::GetFunctionStepValues Method](#)
[ISimulationForceFeatureData::SetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::SetFunctionStepValues Method](#)
[ISimulationForceFeatureData::FunctionConstantValue Property](#)
[ISimulationForceFeatureData::ForceFunctionType Property](#)

Availability

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FunctionInterpolatedValues Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : FunctionInterpolatedValues Property (ISimulationForceFeatureData)

Gets or sets the interpolated values for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Property FunctionInterpolatedValues As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim value As System.Object

instance.FunctionInterpolatedValues = value

value = instance.FunctionInterpolatedValues
```

C#

```
System.Object FunctionInterpolatedValues {get; set;}
```

C++/CLI

```
property System.Object^ FunctionInterpolatedValues {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of interpolated values (time and force)

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::FunctionInterpolatedValues](#).

Remarks

SOLIDWORKS interpolates a spline between the data points.

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::GetInterpolatedValue Method](#)
[ISimulationForceFeatureData::InterpolationScheme Property](#)

Availability

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SOLIDWORKS API Help

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GetEndPoints Method (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : GetEndPoints Method (ISimulationForceFeatureData)

Gets the end points of this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEndPoints( _
    ByRef PDirDisp1 As System.Object, _
    ByRef PDirDisp2 As System.Object, _
    ByRef Type1 As System.Integer, _
    ByRef Type2 As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim PDirDisp1 As System.Object
Dim PDirDisp2 As System.Object
Dim Type1 As System.Integer
Dim Type2 As System.Integer
```

```
Dim value As System.Boolean
```

```
value = instance.GetEndPoints(PDirDisp1, PDirDisp2, Type1, Type2)
```

C#

```
System.bool GetEndPoints(  
    out System.object PDirDisp1,  
    out System.object PDirDisp2,  
    out System.int Type1,  
    out System.int Type2  
)
```

C++/CLI

```
System.bool GetEndPoints(  
& [Out] System.Object^ PDirDisp1,  
& [Out] System.Object^ PDirDisp2,  
& [Out] System.int Type1,  
& [Out] System.int Type2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PDirDisp1

Force end point

PDirDisp2

Force end point

Type1

Type of end point as defined in [swSelectType_e](#)

Type2

Type of end point as defined in [swSelectType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::GetEndPoints](#).

See Also

[ISimulationForceFeatureData Interface](#)

[ISimulationForceFeatureData Members](#)

[ISimulationForceFeatureData::SetEndPoints Method](#)

Availability

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GetFunctionHarmonicValues Method (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : GetFunctionHarmonicValues Method (ISimulationForceFeatureData)

Gets the harmonic function values for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFunctionHarmonicValues( _  
    ByRef Amplitude As System.Double, _  
    ByRef Frequency As System.Double, _  
    ByRef Average As System.Double, _  
    ByRef PhaseShift As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



Copy Code

```

Dim instance As ISimulationForceFeatureData
Dim Amplitude As System.Double
Dim Frequency As System.Double
Dim Average As System.Double
Dim PhaseShift As System.Double
Dim value As System.Boolean

value = instance.GetFunctionHarmonicValues(Amplitude, Frequency, Average, PhaseShift)

```

C#

```

System.bool GetFunctionHarmonicValues(
    out System.double Amplitude,
    out System.double Frequency,
    out System.double Average,
    out System.double PhaseShift
)

```

C++/CLI

```

System.bool GetFunctionHarmonicValues(
    & [Out] System.double Amplitude,
    & [Out] System.double Frequency,
    & [Out] System.double Average,
    & [Out] System.double PhaseShift
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Amplitude

Amplitude of the function measured peak to peak

Frequency

Frequency of the function

Average

Average value of the function; the function oscillates about this value

PhaseShift

Phase shift for the function

Return Value

True if the operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::GetFunctionHarmonicValues](#).

See Also

[ISimulationForceFeatureData Interface](#)

[ISimulationForceFeatureData Members](#)

[ISimulationForceFeatureData::SetFunctionHarmonicValues Method](#)

[ISimulationForceFeatureData::GetFunctionStepValues Method](#)

[ISimulationForceFeatureData::SetFunctionStepValues Method](#)

[ISimulationForceFeatureData::FunctionConstantValue Property](#)

[ISimulationForceFeatureData::FunctionExpression Property](#)

[ISimulationForceFeatureData::ForceFunctionType Property](#)

Availability

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GetFunctionStepValues Method (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : GetFunctionStepValues Method (ISimulationForceFeatureData)

Gets the step function, whose magnitude transitions smoothly from one value to another value, for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFunctionStepValues( _
    ByRef F1InitialValue As System.Double, _
    ByRef T1StartStepTime As System.Double, _
    ByRef F2InitialValue As System.Double, _
    ByRef T2EndStepTime As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim F1InitialValue As System.Double
Dim T1StartStepTime As System.Double
Dim F2InitialValue As System.Double
Dim T2EndStepTime As System.Double
Dim value As System.Boolean

value = instance.GetFunctionStepValues(F1InitialValue, T1StartStepTime, F2InitialValue,
T2EndStepTime)
```

C#

```
System.bool GetFunctionStepValues(
    out System.double F1InitialValue,
    out System.double T1StartStepTime,
    out System.double F2InitialValue,
    out System.double T2EndStepTime
)
```

C++/CLI

```
System.bool GetFunctionStepValues(
    & [Out] System.double F1InitialValue,
    & [Out] System.double T1StartStepTime,
    & [Out] System.double F2InitialValue,
    & [Out] System.double T2EndStepTime
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

F1InitialValue
Value of the function before the step

T1StartStepTime
Time at which the step begins

F2InitialValue
Value of the function after the step

T2EndStepTime
Time at which the step ends

Return Value

True if the operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::GetFunctionStepValues](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::SetFunctionStepValues Method](#)
[ISimulationForceFeatureData::GetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::FunctionConstantValue Property](#)
[ISimulationForceFeatureData::FunctionExpression Property](#)
[ISimulationForceFeatureData::ForceFunctionType Property](#)
[ISimulationForceFeatureData::SetFunctionHarmonicValues Method](#)

Availability

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GetInterpolatedValue Method (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : GetInterpolatedValue Method (ISimulationForceFeatureData)

Gets the interpolated value at the specified time for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Function GetInterpolatedValue( _  
    ByVal Time As System.Double _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim Time As System.Double  
Dim value As System.Double  
  
value = instance.GetInterpolatedValue(Time)
```

C#

```
System.double GetInterpolatedValue(  
    System.double Time  
)
```

C++/CLI

```
System.double GetInterpolatedValue(  
& System.double Time  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Time
Time at which to get the interpolated value

Return Value

Interpolated value

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::GetInterpolatedValue](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::InterpolationScheme Property](#)
[ISimulationForceFeatureData::FunctionInterpolatedValues Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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InterpolationScheme Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : InterpolationScheme Property (ISimulationForceFeatureData)

Gets the interpolation scheme for this Force feature.

.NET Syntax

Visual Basic (Declaration)

Property InterpolationScheme As System.Integer

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim value As System.Integer

instance.InterpolationScheme = value
value = instance.InterpolationScheme
```

C#

```
System.int InterpolationScheme {get; set;}
```

C++/CLI

```
property System.int InterpolationScheme {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Interpolation scheme as defined by [swInterpolationType](#) e

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::InterpolationScheme](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::GetInterpolatedValue Method](#)
[ISimulationForceFeatureData::FunctionInterpolatedValues Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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LoadReferences Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : LoadReferences Property (ISimulationForceFeatureData)

Gets or sets the load references for this Force feature.

.NET Syntax

Visual Basic (Declaration)

Property LoadReferences As System.Object

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim value As System.Object

instance.LoadReferences = value
value = instance.LoadReferences
```

C#

```
System.object LoadReferences {get; set;}
```

C++/CLI

```
property System.Object^ LoadReferences {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Load-bearing [faces](#) or [edges](#)

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::LoadReferences](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

LoadSplineData Method (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : LoadSplineData Method (ISimulationForceFeatureData)

Loads the spline data from the specified file for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Function LoadSplineData(  
    ByVal Value As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim Value As System.String  
Dim value As System.Boolean
```

```
value = instance.LoadSplineData(Value)
```

C#

```
System.bool LoadSplineData(  
    System.string Value  
)
```

C++/CLI

```
System.bool LoadSplineData(  
& System.String^ Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Value

File from which to load the spline data point values

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::LoadSplineData](#).

Remarks

The file should contain one data point per line. The data point consists of two values:

- Time
- Value at that time

Use commas or spaces as separators between the values.

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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ReactionLocation Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : ReactionLocation Property (ISimulationForceFeatureData)

Gets or sets the location at which to apply the force for an action/reaction force.

.NET Syntax

Visual Basic (Declaration)

Property ReactionLocation As System.Object

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim value As System.Object
```

```
instance.ReactionLocation = value
```

```
value = instance.ReactionLocation
```

C#

```
System.object ReactionLocation {get; set;}
```

C++/CLI

```
property System.Object^ ReactionLocation {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Vertex, circle/sphere/torus center, or edge midpoint

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ReactionLocation](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::ActionType Property](#)
[ISimulationForceFeatureData::ActionDirection Property](#)
[ISimulationForceFeatureData::ActionLocation Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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ReferenceComponent Property (ISimulationForceFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : ReferenceComponent Property (ISimulationForceFeatureData)

Gets or sets the component to serve as a reference frame for the force.

Visual Basic Syntax

Visual Basic (Declaration)

Property ReferenceComponent As Component2

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim value As Component2
```

```
instance.ReferenceComponent = value
```

```
value = instance.ReferenceComponent
```

C#

```
Component2 ReferenceComponent {get; set;}
```

C++/CLI

```
property Component2^ ReferenceComponent {  
    Component2^ get();  
    void set ( & Component2^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Component](#) to serve as a reference frame

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ReferenceComponent](#).

Example

[Create Force Feature \(VBA\)](#)

Remarks

Specifying a component to serve as a reference frame is optional. If a component is not specified, then the reference frame is the global ground.

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ReverseDirection Property (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : ReverseDirection Property (ISimulationForceFeatureData)

Gets or sets whether to reverse the direction of the force.

.NET Syntax

Visual Basic (Declaration)

```
Property ReverseDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData  
Dim value As System.Boolean
```

```
instance.ReverseDirection = value
```

```
value = instance.ReverseDirection
```

C#

```
System.bool ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool ReverseDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse the direction of the force, false to not

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::ReverseDirection](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::ActionDirection Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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SetEndPoints Method (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : SetEndPoints Method (ISimulationForceFeatureData)

Sets the end points for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Function SetEndPoints( _  
    ByVal PDirDispl As System.Object, _  
    ByVal PDirDisp2 As System.Object _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationForceFeatureData
Dim PDirDisp1 As System.Object
Dim PDirDisp2 As System.Object
Dim value As System.Boolean
```

```
value = instance.SetEndPoints(PDirDisp1, PDirDisp2)
```

C#

```
System.bool SetEndPoints(
    System.object PDirDisp1,
    System.object PDirDisp2
)
```

C++/CLI

```
System.bool SetEndPoints(
    & System.Object^ PDirDisp1,
    & System.Object^ PDirDisp2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PDirDisp1
Force end point
PDirDisp2
Force end point

Return Value

True if the operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::SetEndPoints](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::GetEndPoints Method](#)

Availability

lidWorks 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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SetFunctionHarmonicValues Method (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : SetFunctionHarmonicValues Method (ISimulationForceFeatureData)

Sets the harmonic function values for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Function SetFunctionHarmonicValues( _
    ByVal Amplitude As System.Double, _
    ByVal Frequency As System.Double, _
    ByVal Average As System.Double, _
    ByVal PhaseShift As System.Double _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ISimulationForceFeatureData
Dim Amplitude As System.Double
```



```
Dim Frequency As System.Double
Dim Average As System.Double
Dim PhaseShift As System.Double
Dim value As System.Boolean
```

```
value = instance.SetFunctionHarmonicValues(Amplitude, Frequency, Average, PhaseShift)
```

C#

```
System.bool SetFunctionHarmonicValues(
    System.double Amplitude,
    System.double Frequency,
    System.double Average,
    System.double PhaseShift
)
```

C++/CLI

```
System.bool SetFunctionHarmonicValues(
    & System.double Amplitude,
    & System.double Frequency,
    & System.double Average,
    & System.double PhaseShift
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Amplitude

Amplitude of the function measured peak to peak

Frequency

Frequency of the function

Average

Average value of the function; the function oscillates about this value

PhaseShift

Phase shift for the function

Return Value

True if the operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::SetFunctionHarmonicValues](#).

See Also

[ISimulationForceFeatureData Interface](#)

[ISimulationForceFeatureData Members](#)

[ISimulationForceFeatureData::GetFunctionHarmonicValues Method](#)

[ISimulationForceFeatureData::GetFunctionStepValues Method](#)

[ISimulationForceFeatureData::SetFunctionStepValues Method](#)

[ISimulationForceFeatureData::FunctionConstantValue Property](#)

[ISimulationForceFeatureData::FunctionExpression Property](#)

[ISimulationForceFeatureData::ForceFunctionType Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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SetFunctionStepValues Method (ISimulationForceFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationForceFeatureData Interface](#) : SetFunctionStepValues Method (ISimulationForceFeatureData)

Sets the step function, whose magnitude transitions smoothly from one value to another value, for this Force feature.

.NET Syntax

Visual Basic (Declaration)

```
Function SetFunctionStepValues( _
    ByVal fInitialValue As System.Double, _
```

```

    ByVal T1StartStepTime As System.Double, _
    ByVal F2InitialValue As System.Double, _
    ByVal T2EndStepTime As System.Double _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As ISimulationForceFeatureData
Dim F1InitialValue As System.Double
Dim T1StartStepTime As System.Double
Dim F2InitialValue As System.Double
Dim T2EndStepTime As System.Double
Dim value As System.Boolean

```

```

value = instance.SetFunctionStepValues(F1InitialValue, T1StartStepTime, F2InitialValue,
T2EndStepTime)

```

C#

```

System.bool SetFunctionStepValues(
    System.double F1InitialValue,
    System.double T1StartStepTime,
    System.double F2InitialValue,
    System.double T2EndStepTime
)

```

C++/CLI

```

System.bool SetFunctionStepValues(
& System.double F1InitialValue,
& System.double T1StartStepTime,
& System.double F2InitialValue,
& System.double T2EndStepTime
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

F1InitialValue
Value of the function before the step

T1StartStepTime
Time at which the step begins

F2InitialValue
Value of the function after the step

T2EndStepTime
Time at which the step ends

Return Value

True if the operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [SimulationForceFeatureData::SetFunctionStepValues](#).

See Also

[ISimulationForceFeatureData Interface](#)
[ISimulationForceFeatureData Members](#)
[ISimulationForceFeatureData::GetFunctionStepValues Method](#)
[ISimulationForceFeatureData::GetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::SetFunctionHarmonicValues Method](#)
[ISimulationForceFeatureData::FunctionConstantValue Property](#)
[ISimulationForceFeatureData::FunctionExpression Property](#)
[ISimulationForceFeatureData::ForceFunctionType Property](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

ISimulationGravityFeatureData (Type)

ISimulationGravityFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationGravityFeatureData Interface

Allows access to a gravity feature in SOLIDWORKS Motion studies.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISimulationGravityFeatureData
```

Visual Basic (Usage)



```
Dim instance As ISimulationGravityFeatureData
```

C#

```
public interface ISimulationGravityFeatureData
```

C++/CLI

```
public interface class ISimulationGravityFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [SimulationGravityFeatureData](#).

Example

[Create Simulation Gravity Feature \(VBA\)](#)
[Create Simulation Gravity Feature \(VB.NET\)](#)
[Create Simulation Gravity Feature \(C#\)](#)

Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#)

[IMotionStudy::CreateDefinition](#)

Access Diagram

[SimulationGravityFeatureData](#)

See Also

[ISimulationGravityFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISimulationMotorFeatureData Interface](#)
[Simulation3DContactFeatureData Interface](#)
[SimulationDamperFeatureData Interface](#)
[SimulationForceFeatureData Interface](#)
[ISimulationSpringFeatureData Interface](#)

ISimulationGravityFeatureData Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationGravityFeatureData Interface

The following tables list the members exposed by [ISimulationGravityFeatureData](#).

Public Properties

Name	Description
 Axis	Gets or sets the axis for this Gravity feature.
 DirectionReference	Gets or sets the direction of in which gravity moves.
 ReverseDirection	Gets or sets whether to reverse the direction of gravity.
 Strength	Gets or sets the gravitational strength.

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 See Also

[ISimulationGravityFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISimulationMotorFeatureData Interface](#)
[Simulation3DContactFeatureData Interface](#)
[SimulationDamperFeatureData Interface](#)
[SimulationForceFeatureData Interface](#)
[ISimulationSpringFeatureData Interface](#)

SOLIDWORKS API Help

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ISimulationGravityFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationGravityFeatureData Interface

For a list of all members of this type, see [ISimulationGravityFeatureData members](#).

 Public Properties

Name	Description
 Axis	Gets or sets the axis for this Gravity feature.
 DirectionReference	Gets or sets the direction of in which gravity moves.
 ReverseDirection	Gets or sets whether to reverse the direction of gravity.
 Strength	Gets or sets the gravitational strength.

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 See Also

[ISimulationGravityFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISimulationMotorFeatureData Interface](#)
[Simulation3DContactFeatureData Interface](#)
[SimulationDamperFeatureData Interface](#)
[SimulationForceFeatureData Interface](#)
[ISimulationSpringFeatureData Interface](#)

SOLIDWORKS API Help

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Axis Property (ISimulationGravityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationGravityFeatureData Interface](#) : Axis Property (ISimulationGravityFeatureData)

Gets or sets the axis for this Gravity feature.

 .NET Syntax

Visual Basic (Declaration)

Property Axis As System.Integer

Visual Basic (Usage)



```
Dim instance As ISimulationGravityFeatureData  
Dim value As System.Integer  
  
instance.Axis = value  
  
value = instance.Axis
```

C#

```
System.int Axis {get; set;}
```

C++/CLI

```
property System.int Axis {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Axis as defined in [swSimulationGravityAxisName_e](#)

Visual Basic for Applications (VBA) Syntax

See [SimulationGravityFeatureData::Axis](#).

Example

See the [ISimulationGravityFeatureData](#) examples.

See Also

[ISimulationGravityFeatureData Interface](#)
[ISimulationGravityFeatureData Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

DirectionReference Property (ISimulationGravityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationGravityFeatureData Interface](#) : DirectionReference Property (ISimulationGravityFeatureData)

Gets or sets the direction of in which gravity moves.

.NET Syntax

Visual Basic (Declaration)

```
Property DirectionReference As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISimulationGravityFeatureData  
Dim value As System.Object  
  
instance.DirectionReference = value  
value = instance.DirectionReference
```

C#

```
System.object DirectionReference {get; set;}
```

C++/CLI

```
property System.Object^ DirectionReference {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Linear edge, planar face, axis, or plane indicating the direction of gravity

Visual Basic for Applications (VBA) Syntax

See [SimulationGravityFeatureData::DirectionReference](#).

See Also

[ISimulationGravityFeatureData Interface](#)
[ISimulationGravityFeatureData Members](#)
[ISimulationGravityFeatureData::ReverseDirection Property](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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ReverseDirection Property (ISimulationGravityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationGravityFeatureData Interface](#) : ReverseDirection Property (ISimulationGravityFeatureData)

Gets or sets whether to reverse the direction of gravity.

.NET Syntax

Visual Basic (Declaration)

Property ReverseDirection As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISimulationGravityFeatureData  
Dim value As System.Boolean
```

```
instance.ReverseDirection = value
```

```
value = instance.ReverseDirection
```

C#

```
System.bool ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool ReverseDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse the direction of gravity, false to not

Visual Basic for Applications (VBA) Syntax

See [SimulationGravityFeatureData::ReverseDirection](#).

Example

See the [ISimulationGravityFeatureData](#) examples.

See Also

[ISimulationGravityFeatureData Interface](#)
[ISimulationGravityFeatureData Members](#)
[ISimulationGravityFeatureData::DirectionReference Property](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Strength Property (ISimulationGravityFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationGravityFeatureData Interface](#) : Strength Property (ISimulationGravityFeatureData)

Gets or sets the gravitational strength.

.NET Syntax

Visual Basic (Declaration)

Property Strength As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulationGravityFeatureData  
Dim value As System.Double
```

```
instance.Strength = value
```

```
value = instance.Strength
```

C#

```
System.double Strength {get; set;}
```

C++/CLI

```
property System.double Strength {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Gravitational strength

Visual Basic for Applications (VBA) Syntax

See [SimulationGravityFeatureData::Strength](#).

Example

See the [ISimulationGravityFeatureData](#) examples.

See Also

[ISimulationGravityFeatureData Interface](#)
[ISimulationGravityFeatureData Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

ISimulationLinearSpringFeatureData (Type)

ISimulationLinearSpringFeatureData Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationLinearSpringFeatureData Interface

Obsolete. Superseded by [ISimulationSpringFeatureData](#).

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISimulationLinearSpringFeatureData
```

Visual Basic (Usage)



```
Dim instance As ISimulationLinearSpringFeatureData
```

C#

```
public interface ISimulationLinearSpringFeatureData
```

C++/CLI

```
public interface class ISimulationLinearSpringFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [SimulationLinearSpringFeatureData](#).

Accessors

[IFeature::GetDefinition](#) and [IFeature::!GetDefinition](#)

[IMotionStudy::CreateFeature](#)

Access Diagram

[SimulationLinearSpringFeatureData](#)

See Also

[ISimulationLinearSpringFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISimulationLinearSpringFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationLinearSpringFeatureData Interface

The following tables list the members exposed by [ISimulationLinearSpringFeatureData](#).

Public Properties

Name	Description
 FreeLength	Obsolete. Superseded by ISimulationSpringFeatureData::FreeLength .
 SpringConstant	Obsolete. Superseded by ISimulationSpringFeatureData::SpringConstant .

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Public Methods

Name	Description
 GetEndPoint	Obsolete. Superseded by ISimulationSpringFeatureData::GetEndPoint .
 SetEndPoint	Obsolete. Superseded by ISimulationSpringFeatureData::SetEndPoint .

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 See Also

[ISimulationLinearSpringFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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ISimulationLinearSpringFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationLinearSpringFeatureData Interface

For a list of all members of this type, see [ISimulationLinearSpringFeatureData members](#).

 Public Methods

Name	Description
 GetEndPoint	Obsolete. Superseded by ISimulationSpringFeatureData::GetEndPoint .
 SetEndPoint	Obsolete. Superseded by ISimulationSpringFeatureData::SetEndPoint .

[Top](#)

 See Also

[ISimulationLinearSpringFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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ISimulationLinearSpringFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationLinearSpringFeatureData Interface

For a list of all members of this type, see [ISimulationLinearSpringFeatureData members](#).

 Public Properties

Name	Description
 FreeLength	Obsolete. Superseded by ISimulationSpringFeatureData::FreeLength .
 SpringConstant	Obsolete. Superseded by ISimulationSpringFeatureData::SpringConstant .

[Top](#)

 See Also

[ISimulationLinearSpringFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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FreeLength Property (ISimulationLinearSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationLinearSpringFeatureData Interface](#) : FreeLength Property (ISimulationLinearSpringFeatureData)

Obsolete. Superseded by [ISimulationSpringFeatureData::FreeLength](#).

 .NET Syntax

Visual Basic (Declaration)

Property FreeLength As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulationLinearSpringFeatureData
Dim value As System.Double

instance.FreeLength = value

value = instance.FreeLength
```

C#

```
System.double FreeLength {get; set;}
```

C++/CLI

```
property System.double FreeLength {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Length

Visual Basic for Applications (VBA) Syntax

See [SimulationLinearSpringFeatureData::FreeLength](#).

See Also

[ISimulationLinearSpringFeatureData Interface](#)
[ISimulationLinearSpringFeatureData Members](#)

Availability

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GetEndPoints Method (ISimulationLinearSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationLinearSpringFeatureData Interface](#) : GetEndPoints Method (ISimulationLinearSpringFeatureData)

Obsolete. Superseded by [ISimulationSpringFeatureData::GetEndPoints](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub GetEndPoints( _
    ByRef PDirDisp1 As System.Object, _
    ByRef PDirDisp2 As System.Object, _
    ByRef Type1 As System.Integer, _
    ByRef Type2 As System.Integer _
)
```

Visual Basic (Usage)



```
Dim instance As ISimulationLinearSpringFeatureData
Dim PDirDisp1 As System.Object
Dim PDirDisp2 As System.Object
Dim Type1 As System.Integer
Dim Type2 As System.Integer
```

```
instance.GetEndPoints(PDirDisp1, PDirDisp2, Type1, Type2)
```

C#

```

void GetEndPoints(
    out System.object PDirDisp1,
    out System.object PDirDisp2,
    out System.int Type1,
    out System.int Type2
)

```

C++/CLI

```

void GetEndPoints(
    & [Out] System.Object^ PDirDisp1,
    & [Out] System.Object^ PDirDisp2,
    & [Out] System.int Type1,
    & [Out] System.int Type2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PDirDisp1

Linear edge, vertex, sketch segment, or sketch point

PDirDisp2

Linear edge, vertex, sketch segment, or sketch point

Type1

Type of end point as defined in [swSelectType_e](#)

Type2

Type of end point as defined in [swSelectType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SimulationLinearSpringFeatureData::GetEndPoints](#).

See Also

[ISimulationLinearSpringFeatureData Interface](#)

[ISimulationLinearSpringFeatureData Members](#)

[ISimulationLinearSpringFeatureData::SetEndPoints Method](#)

Availability

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SOLIDWORKS API Help

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SetEndPoints Method (ISimulationLinearSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationLinearSpringFeatureData Interface](#) : SetEndPoints Method (ISimulationLinearSpringFeatureData)

Obsolete. Superseded by [ISimulationSpringFeatureData::SetEndPoints](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub SetEndPoints( _
    ByVal PDirDisp1 As System.Object, _
    ByVal PDirDisp2 As System.Object _
)

```

Visual Basic (Usage)

```

Dim instance As ISimulationLinearSpringFeatureData
Dim PDirDisp1 As System.Object
Dim PDirDisp2 As System.Object

instance.SetEndPoints(PDirDisp1, PDirDisp2)

```

C#



```
void SetEndPoints(
    System.object PDirDisp1,
    System.object PDirDisp2
)
```

C++/CLI

```
void SetEndPoints(
    & System.Object^ PDirDisp1,
    & System.Object^ PDirDisp2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PDirDisp1
Linear edge, vertex, or sketch point

PDirDisp2
Linear edge, vertex, or sketch point

Visual Basic for Applications (VBA) Syntax

See [SimulationLinearSpringFeatureData::SetEndPoints](#).

See Also

[ISimulationLinearSpringFeatureData Interface](#)
[ISimulationLinearSpringFeatureData Members](#)
[ISimulationLinearSpringFeatureData::GetEndPoints Method](#)

Availability

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SpringConstant Property (ISimulationLinearSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationLinearSpringFeatureData Interface](#) : SpringConstant Property (ISimulationLinearSpringFeatureData)

Obsolete. Superseded by [ISimulationSpringFeatureData::SpringConstant](#).

.NET Syntax

Visual Basic (Declaration)

```
Property SpringConstant As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISimulationLinearSpringFeatureData
Dim value As System.Double
```

```
instance.SpringConstant = value
```

```
value = instance.SpringConstant
```

C#

```
System.double SpringConstant {get; set;}
```

C++/CLI

```
property System.double SpringConstant {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Strength

Visual Basic for Applications (VBA) Syntax

See [SimulationLinearSpringFeatureData::SpringConstant](#).

See Also

[ISimulationLinearSpringFeatureData Interface](#)
[ISimulationLinearSpringFeatureData Members](#)

Availability

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